1 Find the value of 3^{-1} ← flip

$\left(\frac{3}{1}\right)^{-1}$

$\frac{1}{3}$

(Total for question 1 is 1 mark)

2 Find the value of $\left(\frac{4}{5}\right)^{-1}$

$\frac{5}{4}$

(Total for question 2 is 1 mark)

3 Find the value of 5^{-1}

$\frac{1}{5}$

(Total for question 3 is 1 mark)

4 Find the reciprocal of 3

$\frac{1}{3}$

(Total for question 4 is 1 mark)

5 Find the value of $100^{\frac{1}{2}}$ ← square root

10

(Total for question 5 is 1 mark)

6 Find the value of $64^{\frac{1}{2}}$

8

(Total for question 6 is 1 mark)

7 Find the value of $49^{\frac{1}{2}}$

7

(Total for question 7 is 1 mark)

8 Find the value of $81^{\frac{1}{2}}$

9

(Total for question 8 is 1 mark)

9 Find the value of $36^{-\frac{1}{2}}$ ← square root and flip

$\frac{1}{6}$

(Total for question 9 is 1 mark)

10 Find the value of $64^{\frac{1}{3}}$ ← cube root

4

(Total for question 10 is 1 mark)

11 Find the value of $8^{\frac{1}{3}}$

2

(Total for question 11 is 1 mark)

12 Find the value of $27^{\frac{1}{3}}$

3

(Total for question 12 is 1 mark)

13 Find the value of $125^{\frac{1}{3}}$

5

(Total for question 13 is 1 mark)

14 Find the value of $64^{-\frac{1}{3}}$ ← cube root and flip

$\frac{1}{4}$

(Total for question 14 is 1 mark)

15 Find the value of $64^{-\frac{2}{3}}$ ← cube root, square and flip

$(4)^{-2}$

$(16)^{-1}$

$\frac{1}{16}$

(Total for question 15 is 2 marks)

16 Find the value of $125^{\frac{2}{3}}$ ← cube root and square

$(5)^{2}$

25

(Total for question 16 is 2 marks)

17 Find the value of $8^{-\frac{2}{3}}$

$(2)^{-2}$

$(4)^{-1}$

$\frac{1}{4}$

(Total for question 17 is 2 marks)

18 Find the value of $27^{-\frac{2}{3}}$

$(3)^{-2}$

$(9)^{-1}$

$\frac{1}{9}$

(Total for question 18 is 2 marks)

19 Find the value of $(8x^{6})^{\frac{2}{3}}$

$(2x^{2})^{2}$

$4x^{4}$

(Total for question 19 is 2 marks)

20 Find the value of $\left(\frac{64}{125}\right)^{-\frac{2}{3}}$

$\left(\frac{4}{5}\right)^{-2}$

$\left(\frac{16}{25}\right)^{-1}$

$\frac{25}{16}$

(Total for question 20 is 2 marks)

21 Find the value of $\left(\frac{25}{16}\right)^{-\frac{3}{2}}$

$\left(\frac{5}{4}\right)^{-3}$

$\left(\frac{125}{64}\right)^{-1}$

$\frac{64}{125}$

(Total for question 21 is 2 marks)

22 Find the value of $\left(\frac{8}{27}\right)^{-\frac{2}{3}}$

$\left(\frac{2}{3}\right)^{-2}$

$\left(\frac{4}{9}\right)^{-1}$

$\frac{9}{4}$

(Total for question 22 is 2 marks)

23 Find the value of $\left(\frac{9}{4}\right)^{-\frac{3}{2}}$

$\left(\frac{3}{2}\right)^{-3}$

$\left(\frac{27}{8}\right)^{-1}$

$\frac{8}{27}$

(Total for question 23 is 2 marks)

24 Find the value of $\sqrt[4]{2 \times 8 \times 10^{12}}$

$\sqrt[4]{16 \times 10^{12}}$

2×10^{3}

2000

(Total for question 24 is 2 marks)

25 Find the value of $\sqrt[3]{4 \times 16 \times 10^{15}}$

$\sqrt[3]{64 \times 10^{15}}$

4×10^{5}

400000

(Total for question 25 is 2 marks)

26 Given that $3 \times \sqrt{3} = 3^n$
Find the value n.

$3^1 \times 3^{\frac{1}{2}} = 3^{\frac{3}{2}}$

$\frac{3}{2}$

(Total for question 26 is 2 marks)

27 Given that $3 \times \sqrt{27} = 3^n$
Find the value n.

$3^1 \times \sqrt{3^3}$

$3^1 \times 3^{\frac{3}{2}} = 3^{\frac{5}{2}}$

$\frac{5}{2}$

(Total for question 27 is 2 marks)

28 Given that $x = 2^p$ and $y = 2^q$
Express in terms of x and/or y,

(i) 2^{p+q}

$2^p \times 2^q$

$x \times y$

(ii) 2^{2p}

$2^p \times 2^p$

x^2

(iii) 2^{q-1}

$2^q \div 2^1$

$y/2$

(Total for question 28 is 3 marks)

29 Given that $3^{-n} = 0.2$
Find the value of $(3^n)^2$

$3^{-n} = \frac{1}{5}$

$3^n = 5$

$(3^n)^2 = 25$

25

(Total for question 29 is 2 marks)

30 Given that $5^{-n} = 0.5$
Find the value of $(5^n)^3$

$5^{-n} = \frac{1}{2}$

$5^n = 2$

$(5^n)^3 = 8$

8

(Total for question 30 is 2 marks)

31 Given that $4^n = 8$
Find the value of n.

$4^n = 8$

$(2^2)^n = 8$

$2^{2n} = 2^3$

$2n = 3$

$n = 1.5$

(Total for question 31 is 2 marks)

32 Given that $4^{-n} = 32$
Find the value of n.

$4^{-n} = 32$

$2^{-2n} = 2^5$

$-2n = 5$

$n = -2.5$

$n = -2.5$

(Total for question 32 is 2 marks)